

EC-231 Operating Systems

BS(CE)-2021

LAB REPORT # 2



Submitted By:

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	Presentation (Format (0.5) + Title (0.5) + Conclusion(1) + Procedure(2) + Objective(1))	Calculation/ Coding	Observation/ Program Results	Total
Total	5	5	5	15
Obtained				

**Lab Instructor.
Engr. Fasih Ahmad**

Experiment #2

Linux Shell commands for file operations and their rights

Objective:

The main objectives of this lab are to understand:

1. How to view or edit a text file in Linux.
2. How we can concatenate two or more files into a single file,
3. How to sort contents of file
4. How to display specific lines of a file
5. How to assign different permissions to the file using commands in Ubuntu operating system.

Software Used:

1. Ubuntu
2. Virtual box

Procedure:

1.ls:

Use the `ls` command to list files and directories in the current directory. It provides information about file names, sizes, and permissions.

2.chmod:

The `chmod` command is used to change the permissions of files and directories. It allows you to specify who can read, write, or execute a file.

3. chown:

With the `chown` command, you can change the ownership of a file or directory. This command is often used to transfer ownership between users or groups.

4. cp:

Use the `cp` command to copy files and directories from one location to another. It creates a duplicate of the file or directory with the specified permissions.

5.mv:

The `mv` command is used to move files and directories from one location to another. It also allows you to rename files and directories.

6.rm:

Use the `rm` command to remove files and directories from the system. Be cautious when using this command, as it permanently deletes the specified files.

7. mkdir:

The `mkdir` command is used to create new directories. You can specify the name and permissions of the directory to be created.

8. rmdir:

Use the ``rmdir`` command to remove empty directories from the system. Similar to the ``rm`` command, be careful when using ``rmdir`` as it deletes directories permanently.

Understanding file permissions:

1. In Linux, file permissions are represented by three sets of characters: user, group, and others.
2. Each set can have three permissions: read (r), write (w), and execute (x).
3. The ``ls -l`` command displays file permissions in the format ``drwxr-xr-x``, where the first character indicates the file type, followed by three sets of permissions for user, group, and others.

Understanding file ownership:

1. Each file and directory in Linux has an owner and a group associated with it.
2. The ``chown`` command allows you to change both the owner and the group of a file or directory.

These commands and concepts form the foundation for managing files and directories in a Linux environment, enabling users to control access and organization effectively.

Task

Task 1:

Show Simulation results after performing the following tasks by using suitable commands in Linux. Also show command/commands used for specific task

- a. Create 3 empty text files and name them as `students.txt`, `pstudent.txt` and `fstudents.txt`.
- b. Assign permissions to read only permission to the user.
- c. Enter at least 10 10 different students' names in `pstudent.txt` and `fstudent.txt`. If any error appears, debug that error for user only.
- d. Now create directory named HITEC and assign permissions of read only to user and group.
- e. Copy all files to directory HITEC. If any error appears, debug that error for user and group only.
- f. After that append the file `students.txt` with first five sorted names from `pstudent.txt` and last five sorted names from `fstudent.txt`.
- g. Show the contents of sorted names from file `students.txt`.
- h. Change the permissions of file `students.txt` read only and both other files read and write only.
- i. Now append another student name in `student.txt` file.
- j. Display contents of all the files.
- k. Show permissions and all of the file contents of HITEC.

Stimulation:

```
huzaifashahbaz@huzaifashahbaz-VirtualBox:~$ touch students.txt pstudents.txt fstudents.txt
huzaifashahbaz@huzaifashahbaz-VirtualBox:~$ chmod u=r students.txt
huzaifashahbaz@huzaifashahbaz-VirtualBox:~$ chmod u=r pstudents.txt fstudents.txt
huzaifashahbaz@huzaifashahbaz-VirtualBox:~$ ls -l
total 72
drwxrwxr-x 19 huzaifashahbaz huzaifashahbaz 4096 22:10 26 فروی 'ALL PROGRAMMING LANGUAGES'
drwxrwxrwx 3 huzaifashahbaz huzaifashahbaz 4096 18:35 22 فروی Android
drwxrwxr-x 2 huzaifashahbaz huzaifashahbaz 4096 18:45 18 فروی copy_command
drwxr-xr-x 2 huzaifashahbaz huzaifashahbaz 4096 19:07 3 مارچ Desktop
drwxr-xr-x 2 huzaifashahbaz huzaifashahbaz 4096 13:35 4 مارچ Documents
drwxr-xr-x 2 huzaifashahbaz huzaifashahbaz 4096 00:44 3 مارچ Downloads
drwxrwxr-x 2 huzaifashahbaz huzaifashahbaz 4096 14:06 4 مارچ folder
-rw-rw-r-- 1 huzaifashahbaz huzaifashahbaz 0 14:03 4 مارچ folder.sh
-r--rw-r-- 1 huzaifashahbaz huzaifashahbaz 0 20:18 4 مارچ fstudents.txt
drwxrwxr-x 2 huzaifashahbaz huzaifashahbaz 4096 13:52 4 مارچ huzaifa
drwxrwxr-x 2 huzaifashahbaz huzaifashahbaz 4096 20:15 4 مارچ huzaifa786
drwxrwxr-x 2 huzaifashahbaz huzaifashahbaz 4096 19:18 18 فروی LINUX
drwxr-xr-x 3 huzaifashahbaz huzaifashahbaz 4096 14:06 26 فروی Music
drwxr-xr-x 3 huzaifashahbaz huzaifashahbaz 4096 12:50 4 مارچ Pictures
-r--rw-r-- 1 huzaifashahbaz huzaifashahbaz 0 20:18 4 مارچ pstudents.txt
drwxr-xr-x 2 huzaifashahbaz huzaifashahbaz 4096 14:48 18 فروی Public
drwx----- 9 huzaifashahbaz huzaifashahbaz 4096 14:26 26 فروی snap
-r--rw-r-- 1 huzaifashahbaz huzaifashahbaz 0 20:18 4 مارچ students.txt
drwxr-xr-x 2 huzaifashahbaz huzaifashahbaz 4096 14:48 18 فروی Templates
drwxrwxr-x 2 huzaifashahbaz huzaifashahbaz 4096 13:51 4 مارچ testlab03
drwxrwxr-x 2 huzaifashahbaz huzaifashahbaz 4096 13:30 4 مارچ testlab3
drwxr-xr-x 2 huzaifashahbaz huzaifashahbaz 4096 14:48 18 فروی Videos
huzaifashahbaz@huzaifashahbaz-VirtualBox:~$ echo "student1" >> pstudents.txt
bash: pstudents.txt: Permission denied
huzaifashahbaz@huzaifashahbaz-VirtualBox:~$ echo "student1" >> fstudents.txt
bash: fstudents.txt: Permission denied
huzaifashahbaz@huzaifashahbaz-VirtualBox:~$ ls -l pstudents.txt fstudents.txt
-r--rw-r-- 1 huzaifashahbaz huzaifashahbaz 0 20:18 4 مارچ fstudents.txt
-r--rw-r-- 1 huzaifashahbaz huzaifashahbaz 0 20:18 4 مارچ pstudents.txt
huzaifashahbaz@huzaifashahbaz-VirtualBox:~$ ls -l pstudents.txt fstudents.txt
-r--rw-r-- 1 huzaifashahbaz huzaifashahbaz 0 20:18 4 مارچ fstudents.txt
-r--rw-r-- 1 huzaifashahbaz huzaifashahbaz 0 20:18 4 مارچ pstudents.txt
huzaifashahbaz@huzaifashahbaz-VirtualBox:~$ chmod u+w pstudents.txt fstudents.txt
huzaifashahbaz@huzaifashahbaz-VirtualBox:~$ ls -l pstudents.txt fstudents.txt
-rw-rw-r-- 1 huzaifashahbaz huzaifashahbaz 0 20:18 4 مارچ fstudents.txt
-rw-rw-r-- 1 huzaifashahbaz huzaifashahbaz 0 20:18 4 مارچ pstudents.txt
huzaifashahbaz@huzaifashahbaz-VirtualBox:~$ echo "student2" >> fstudents.txt pstudents.txt
huzaifashahbaz@huzaifashahbaz-VirtualBox:~$ echo "student3" >> fstudents.txt pstudents.txt
huzaifashahbaz@huzaifashahbaz-VirtualBox:~$ echo "student4" >> fstudents.txt pstudents.txt
bash: fstudents.txt: Permission denied
huzaifashahbaz@huzaifashahbaz-VirtualBox:~$ echo "student4" >> fstudents.txt pstudents.txt
huzaifashahbaz@huzaifashahbaz-VirtualBox:~$ echo "student5" >> fstudents.txt pstudents.txt
huzaifashahbaz@huzaifashahbaz-VirtualBox:~$ echo "student7" >> fstudents.txt pstudents.txt
huzaifashahbaz@huzaifashahbaz-VirtualBox:~$ echo "student8" >> fstudents.txt pstudents.txt
huzaifashahbaz@huzaifashahbaz-VirtualBox:~$ echo "student9" >> fstudents.txt pstudents.txt
huzaifashahbaz@huzaifashahbaz-VirtualBox:~$ echo "Rafay" >> fstudents.txt
huzaifashahbaz@huzaifashahbaz-VirtualBox:~$ echo "Tamoor" >> pstudents.txt
huzaifashahbaz@huzaifashahbaz-VirtualBox:~$ mkdir hitec
huzaifashahbaz@huzaifashahbaz-VirtualBox:~$ cp students.txt pstudents.txt fstudents.txt/home/huzaifa/hitec
```

```

huzaifashahbaz@huzaifashahbaz-VirtualBox:~$ cp students.txt pstudents.txt fstudents.txt /home/huzaifashahbaz/hitec
huzaifashahbaz@huzaifashahbaz-VirtualBox:~$ sort pstudents.txt | head -n 5 >> students.txt
bash: students.txt: Permission denied
huzaifashahbaz@huzaifashahbaz-VirtualBox:~$ sort fstudents.txt | tail -n 5 >> students.txt
bash: students.txt: Permission denied
huzaifashahbaz@huzaifashahbaz-VirtualBox:~$ chmod 777 students.txt
huzaifashahbaz@huzaifashahbaz-VirtualBox:~$ sort pstudents.txt | head -n 5 >> students.txt
huzaifashahbaz@huzaifashahbaz-VirtualBox:~$ sort fstudents.txt | head -n 5 >> students.txt
huzaifashahbaz@huzaifashahbaz-VirtualBox:~$ sort students.txt | cat
Rafay
student2 pstudents.txt
student3 pstudents.txt
student4 pstudents.txt
student4 pstudents.txt
Tamloor
huzaifashahbaz@huzaifashahbaz-VirtualBox:~$ chmod u=rw,go=rw pstudents.txt fstudents.txt
huzaifashahbaz@huzaifashahbaz-VirtualBox:~$ chmod u+w students.txt
huzaifashahbaz@huzaifashahbaz-VirtualBox:~$ echo "new student" >> students.txt
huzaifashahbaz@huzaifashahbaz-VirtualBox:~$ chmod u-w students.txt

huzaifashahbaz@huzaifashahbaz-VirtualBox:~$ cat students.txt
Tamloor
Rafay
student2 pstudents.txt
student3 pstudents.txt
student4 pstudents.txt
student4 pstudents.txt
new student
huzaifashahbaz@huzaifashahbaz-VirtualBox:~$ ls -l hitec
total 8
-rw-rw-r-- 1 huzaifashahbaz huzaifashahbaz 190 21:08 4 ماج fstudents.txt
-rw-rw-r-- 1 huzaifashahbaz huzaifashahbaz 8 21:08 4 ماج pstudents.txt
-r--rw-r-- 1 huzaifashahbaz huzaifashahbaz 0 21:08 4 ماج students.txt
huzaifashahbaz@huzaifashahbaz-VirtualBox:~$ cat hitec/*
student2 pstudents.txt
student3 pstudents.txt
student4 pstudents.txt
student4 pstudents.txt
student5 pstudents.txt
student7 pstudents.txt
student8 pstudents.txt
student9 pstudents.txt
Rafay
Tamloor

```

Conclusion:

In conclusion, Linux shell commands allow for efficient file operations such as creating, copying, relocating, and deleting files, as well as managing file permissions with `chmod` and `chown`. These commands allow users to effectively alter files, directories, and permissions, which improves system security and organization. Understanding and using these commands is required for effective file management and system administration in Linux systems.